

CHETUMAL INTL, Mexico

WMO: 767500

Lat: 18.5006N

Lon: 88.3278W

Elev: 30

StdP: 14.68

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	59.8	63.1	55.3	65.1	64.5	57.8	71.5	66.4	23.2	79.2	22.7	79.9	4.1	330	0.370

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.8	93.8	80.5	92.7	80.4	91.5	80.1	82.8	90.2	81.9	89.4	81.5	88.8	7.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.8	160.2	88.1	80.1	156.2	87.4	79.2	151.4	86.4	46.8	90.3	45.8	89.7	45.2	89.4	91.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.1	22.5	21.3	53.8	99.2	2.2	1.9	52.2	100.5	51.0	101.7	49.7	102.8	48.1	104.2		
(5)			52.6	86.9	2.3	1.9	51.0	88.2	49.6	89.4	48.3	90.4	46.6	91.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	75.7	78.3	79.9	83.1	84.7	84.2	84.7	85.1	84.2	81.8	78.5	77.2
	DBStd	4.50	3.95	3.96	3.86	2.83	2.36	2.59	2.56	2.55	2.48	3.01	3.06	3.77
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11487	798	791	927	993	1076	1026	1077	1088	1026	985	856	844
	CDD65	6013	333	372	462	543	611	576	612	623	576	520	406	379
	CDH74	61114	2232	3106	4249	5773	7063	6615	7232	7406	6465	5024	3212	2737
	CDH80	22951	490	821	1224	2052	2949	2687	3122	3345	2732	1846	960	723
Wind	WSAvg	8.1	7.2	8.6	9.7	10.8	10.4	9.2	8.3	8.1	6.9	6.2	5.7	6.0
Precipitation	PrecAvg	51.4	2.6	1.2	1.4	1.5	4.2	7.8	4.9	6.4	7.0	7.7	4.1	2.6
	PrecMax	74.3	7.8	3.3	4.1	5.5	13.2	14.8	13.7	19.4	15.1	23.3	9.4	7.2
	PrecMin	37.4	0.6	0.0	0.2	0.0	0.0	2.3	0.4	1.7	2.2	1.3	0.7	0.4
	PrecStd	8.7	1.7	0.7	1.0	1.2	3.3	3.7	3.2	4.4	3.3	5.1	2.3	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.1	90.0	91.1	92.7	94.0	93.5	94.5	95.7	94.7	93.2	90.6	89.2
		MCWB	77.0	76.6	78.0	79.4	80.2	80.5	80.4	80.5	81.0	80.4	78.5	78.0
	2%	DB	86.2	88.0	89.2	90.9	92.2	91.8	92.9	93.6	92.9	91.2	88.5	87.5
		MCWB	77.1	77.7	77.4	78.9	79.9	80.7	80.4	80.7	80.8	79.8	78.3	77.6
	5%	DB	84.5	86.3	87.6	89.5	90.8	90.0	91.2	91.7	91.1	89.4	87.2	85.9
		MCWB	76.3	77.2	77.4	78.6	79.6	80.5	80.1	80.7	80.7	79.6	78.0	77.3
	10%	DB	82.8	84.6	85.9	87.8	89.3	88.7	89.6	89.9	89.5	87.8	85.8	84.2
		MCWB	75.7	76.2	76.9	78.6	79.5	80.2	79.9	80.3	80.4	79.7	77.6	76.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.6	80.2	80.9	82.1	83.4	83.3	82.8	83.6	83.7	82.7	81.0	80.3
		MCDB	85.1	86.9	88.2	89.4	90.7	89.8	90.8	91.5	91.0	89.4	87.8	86.7
	2%	WB	78.3	79.2	79.6	80.8	81.8	81.9	81.8	82.2	82.3	81.5	80.0	79.2
		MCDB	84.0	85.6	86.6	88.1	89.2	88.8	89.8	90.2	89.8	88.1	86.2	85.2
	5%	WB	77.5	78.2	78.7	79.9	80.9	81.2	81.0	81.5	81.5	80.6	79.2	78.3
		MCDB	83.2	84.4	85.3	87.0	88.2	88.0	88.7	89.3	88.8	87.2	85.1	83.9
	10%	WB	76.6	77.3	77.8	79.1	80.1	80.5	80.4	80.8	80.7	79.9	78.3	77.4
		MCDB	82.0	83.5	84.3	86.0	87.1	87.3	87.7	88.4	87.8	85.9	83.8	82.8
Mean Daily Temperature Range	5% DB	MDBR	15.9	15.7	15.1	13.5	12.6	11.8	12.9	13.8	14.0	14.3	15.8	15.6
		MCDBR	16.2	15.5	15.1	13.8	13.3	12.8	14.0	14.9	15.0	15.5	17.0	16.2
		MCWBR	9.7	8.6	8.1	6.9	6.4	5.8	5.9	6.6	6.6	7.5	9.3	9.1
	5% WB	MCDBR	14.8	14.5	13.8	12.9	12.7	12.2	13.4	14.4	14.3	14.6	15.2	15.1
		MCWBR	9.4	8.7	8.2	7.2	6.5	6.2	6.2	6.8	6.8	7.5	9.1	9.1
Clear-Sky Solar Irradiance	taub	0.369	0.381	0.412	0.459	0.483	0.467	0.495	0.468	0.439	0.411	0.389	0.382	
	taud	2.551	2.500	2.397	2.258	2.220	2.300	2.185	2.283	2.376	2.472	2.503	2.522	
	Ebn at Noon	291	292	285	271	261	263	256	265	273	279	281	282	
	Edh at Noon	31	34	39	45	46	42	47	43	39	34	32	31	
All-Sky Solar Radiation	RadAvg	1309	1557	1785	1926	1809	1631	1743	1733	1605	1474	1375	1262	
	RadStd	90	91	102	88	128	127	131	106	116	106	77	87	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page